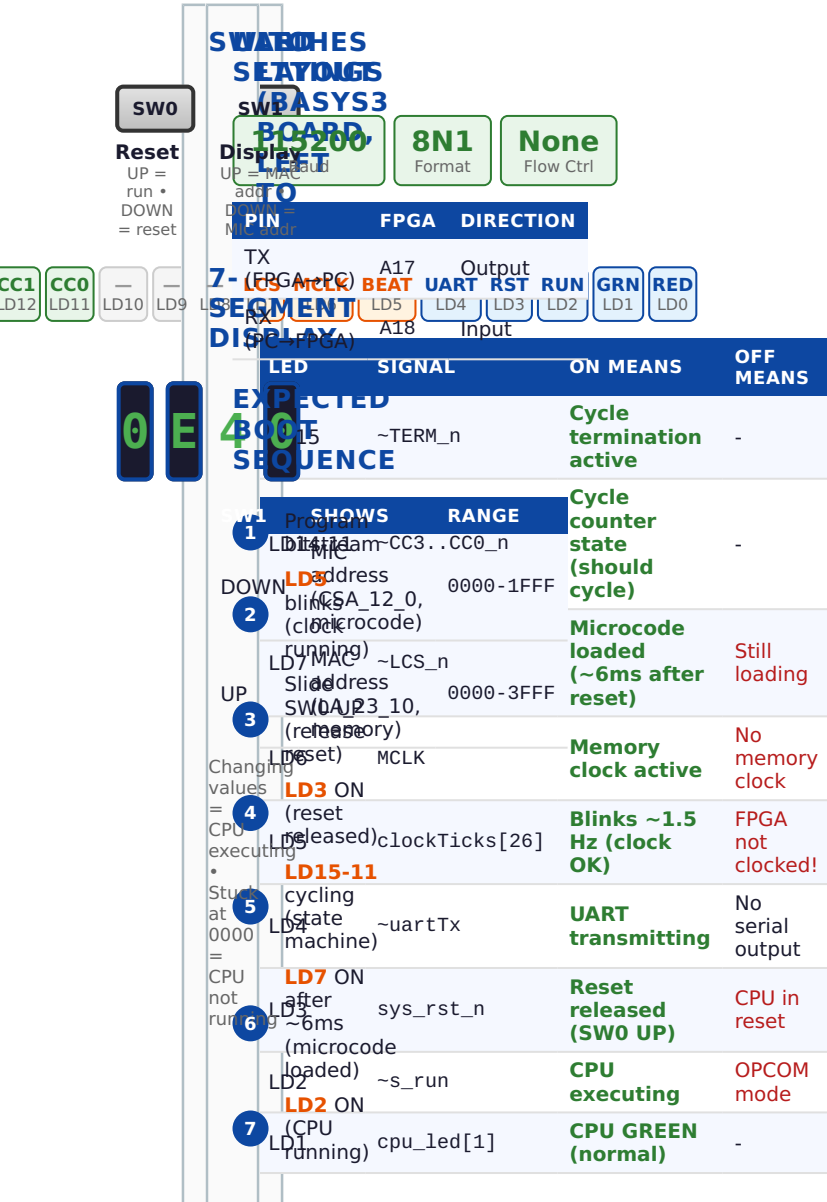


ND-120 FPGA Debug Reference Card

Basys3 (xc7a35tcpg236-1) • Branch: redo-idb • 100 MHz Clock • 115200 Baud



7-seg LED shows changing hex values	cpu_led[0]	CPU RED (error)	Normal
Terminal shows # prompt			

VIVADO ILA DEBUG SIGNALS (MARK_DEBUG)

SIGNAL	WIDTH	LOCATION	DESCRIPTION
s_debug_csa[12:0]	13	ND120_TOP	MIC address (microcode fetch address)
s_debug_la_23_10[13:0]	14	ND120_TOP	MAC address upper (logical address bits 23:10)
s_debug_ca_9_0[9:0]	10	ND120_TOP	MAC address lower (cache address bits 9:0)
s_debug_cc_term[4:0]	5	ND120_TOP	{TERM_n, CC3_n, CC2_n, CC1_n, CC0_n}
s_debug_mclk	1	ND120_TOP	Memory clock
s_debug_lcs_n	1	ND120_TOP	LCS_n (0=loading microcode, 1=loaded)
s_run	1	ND120_TOP	CPU run state (0=running)
s_debug_uartTx	1	ND120_TOP	UART TX line
s_debug_uartRx	1	ND120_TOP	UART RX line
s_debug_cpu_led[6:0]	7	ND120_TOP	CPU status LEDs (internal)

**DEEP
HIERARCHY
SIGNALS
(ADD
VIA
VIVADO
SET
UP
DEBUG)**

HIERARCHICAL PATH	WIDTH	DESCRIPTION
CPU_BOARD/CPU/PROC/ CGA/DELILAH/ s_FIDBO_15_0	16	CGA internal data bus output (FIDBO)
CPU_BOARD/CPU/PROC/ CGA/DELILAH/ s_FIDBI_15_0	16	CGA internal data bus input (FIDBI)
CPU_BOARD/CPU/CS/ s_csbits[63:0]	64	TOPCSB - full 64-bit microcode word
CPU_BOARD/CPU/PROC/ s_idb_erf_out[15:0]	16	Register file output to IDB
CPU_BOARD/I0/UART/ CHIP_32H/txState	3	UART TX state machine
CPU_BOARD/I0/UART/ CHIP_32H/rxState	3	UART RX state machine
CPU_BOARD/CYC/ s_cc0_n	1	Cycle counter bit 0 (inside CYC_36)